

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 28)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops

☐

B Steam washing

☐

C Inflatable booms

☐

D Polymerising agents

☐

0 1 . 2

Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

.....

.....

.....

.....

0 1 . 3

Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

.....

.....

.....

.....

5

0 2 . 1

Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]

Table 1

Method	How it reduces erosion	
Terracing		
Tied ridging		
Windbreaks / hedgerows		
Mulching		
Strip cropping		

.....

0	2	.	2
---	---	---	---

Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

0	2	.	3
---	---	---	---

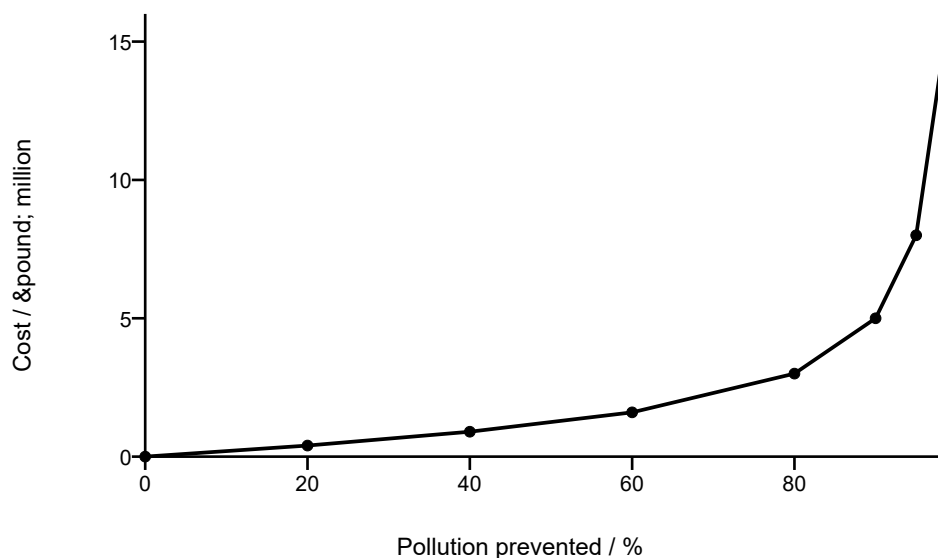
Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

10

03

Figure 1 shows the relationship between the cost of pollution control equipment at a factory and the percentage of a pollutant emission that is prevented.

Figure 1**03****1**

Describe the relationship shown in Figure 1.

[2 marks]**03****2**

Use Figure 1 to calculate the extra cost of increasing the pollution prevented from 90% to 99%.

[1 mark]**Answer**

&pound; million

03**3**

Explain what is meant by the terms **ALARA** and **BATNEEC** and how they would be applied by the factory.

[4 marks]

0 3 . 4

Suggest why the control methods used originally by a factory may no longer be sufficient after several years.

[3 marks]

10

0 4 . 1

Give **three** sources of unburnt hydrocarbons in urban air.

[3 marks]

1

2

3

0 4 . 2

Describe **two** methods used to control hydrocarbon emissions at petrol filling stations.

[2 marks]

1

2

0 4 . 3 Explain how a catalytic converter controls the emissions of carbon monoxide, hydrocarbons and NO_x.

[3 marks]

0 4 . 4 Explain why carbon monoxide is toxic to humans.

[2 marks]

10

0 5 Table 1 shows features of three rechargeable battery types.

Table 1

Battery type	Energy density / Wh kg ⁻¹	Charge-discharge cycles	Storage efficiency / %	Cost per kWh / £	Safety issues
Lead-acid	35	500	80	100	Toxic lead, acid
Nickel-cadmium	50	1500	75	250	Toxic cadmium
Lithium-ion	200	2000	92	150	Fire risk

05 . 1

Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

05 . 2

Give **two** other factors that affect the viability of a rechargeable battery.

[2 marks]

1

.....

2

.....

05 . 3

Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

[2 marks]

.....

.....

.....

05 . 4

Explain how Power-to-Gas (P2G) systems use existing infrastructure.

[2 marks]

.....

.....

.....

.....

10

0	6
---	---

1

[5 marks]

2

[3 marks]

3

[2 marks]

10

07 . 1

Describe the two processes that drive the North Atlantic Conveyor.

[2 marks]

07 . 2

Explain how global climate change could weaken the North Atlantic Conveyor and describe the possible consequence for the climate of the UK.

[4 marks]

07 . 3

Explain why the effects of climate change on UK temperatures are difficult to predict.

[3 marks]

07 . 4

Describe how deep ocean currents are monitored.

[1 mark]

10

08 . 1

Explain how each of the following features of vehicle design reduces energy use.

[6 marks]

Aerodynamics

.....

.....

.....

.....

Reduced mass

.....

.....

.....

.....

Regenerative braking (KERS)

.....

.....

.....

.....

08 . 2

Explain why solid steel wheels on rails lose less energy than pneumatic tyres on roads.

[2 marks]

.....

.....

.....

.....

08 . 3

Describe **two** ways combustion efficiency in an internal combustion engine can be improved.

[2 marks]

1

.....

2

.....

10

09 . 1

Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station

☐

B Tropospheric ozone

☐

C Carbon monoxide from a car exhaust

☐

D Nitrogen monoxide from a diesel engine

☐

09 . 2

Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

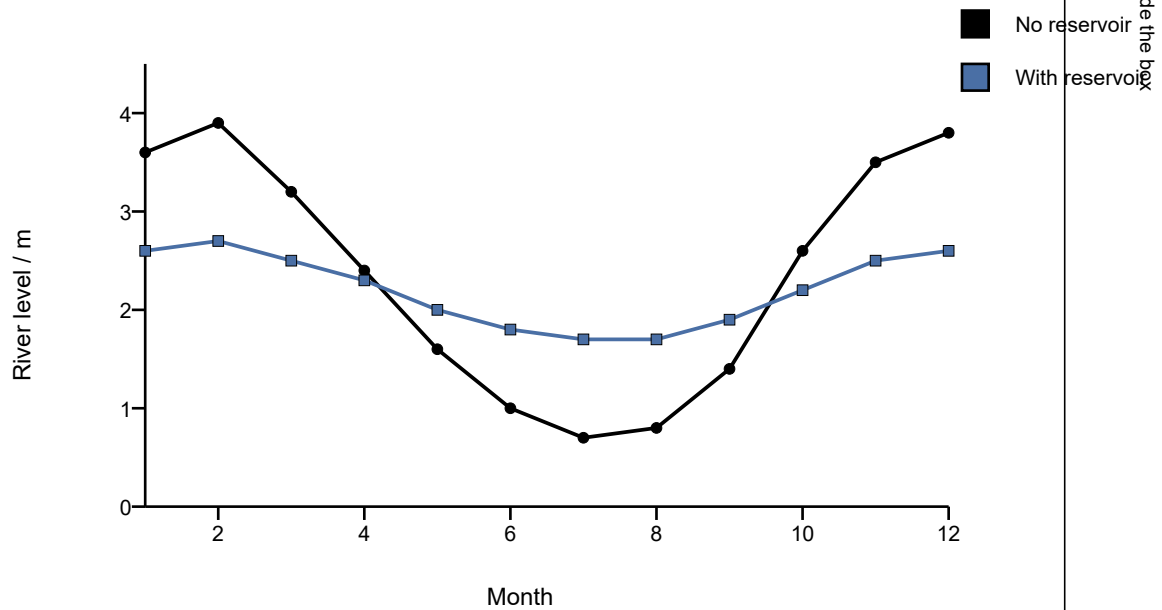
.....

5

1 0

Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0

1

Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer

 m

1 0

2

Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

.....

.....

.....

.....

1 0

3

Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

.....

10 . 4

Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

.....

.....

.....

.....

.....

.....

.....

Extra space

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

15

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
---	---	---	---

Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

END OF QUESTIONS